

JETT KAFKA

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Portfolio: <https://jettk14.github.io/engineering-portfolio/>

PROFILE

Electrical engineering student at McGill University (eligible for internship via McGill EIP) with a 4.0 CGPA and a constant drive to take on and solve harder problems. Built engineering skills beyond the classroom through 13 independent hardware projects and McGill Rocket Team avionics membership; gained hands-on experience in PCB design, embedded systems, hardware testing, and debugging. Beyond engineering, brings 3 summers of experience as a hockey instructor and camp counsellor, developing the communication, leadership, and judgment needed to mentor and manage group dynamics. Seeking a Summer 2027 internship in embedded systems, hardware design, or firmware engineering.

EDUCATION

McGill University <i>Bachelor of Engineering, Electrical Engineering</i>	Montreal, QC 2025 - Present
Member of the McGill Rocket Team Avionics Radio Division	

ENGINEERING EXPERIENCE & PROJECTS

McGill Rocket Team <i>Avionics Radio Division</i>	Montreal, QC 2025 - Present
- Assembled and debugged the SRADIO PCB, a LoRa-based radio board using the SX1262 transceiver for rocket-to-ground communication. - Interpreted component datasheets to ensure correct PCB assembly and operating conditions, preventing component damage and hardware faults during bring-up. - Collaborated in a two-person project team, contributing to technical decisions, troubleshooting, and project execution.	

4-Bit Computer (Independent Project)

- Built a breadboard-based 4-bit computer integrating a program counter, EEPROM program ROM, control logic, RAM, registers, and ALU.
- Debugged timing and synchronization across digital ICs while implementing low-level computer architecture and control logic.

2D Servo LiDAR Mapper (Independent Project)

- Developed a 2D scanning system using a TF Mini Plus LiDAR sensor, closed-loop servos, and Arduino-based control to map distance measurements across a field of view.
- Progressed the system from breadboard prototype through PCB implementation, testing, and final integration.
- Designed and 3D printed a custom enclosure and mounting hardware in Fusion 360.

Successive Approximation Register ADC (Independent Project)

- Built and tested a successive-approximation ADC using digital control logic, an R/2R ladder DAC, comparator feedback, and clocked registers.
- Implemented manual and continuous conversion modes to evaluate ADC behavior under varying analog inputs.

ADDITIONAL EXPERIENCE

Solly's Craft Soda <i>Brand Ambassador</i>	Toronto, ON July - August 2026
- Learned the impact of body language on sales performance, constantly evaluated customer reactions to adjust the sales approach and make more sales. - Interacted with 4000+ potential customers. - Promoted products through in-store sampling, customer engagement and product education.	

Camp Manitou <i>Ball Hockey Instructor and Cabin Counsellor</i>	Parry Sound, ON Part Time June - August 2024 - 2026
- Learned coordination, communication and conflict resolution skills through interacting with 12-13 kids and ensuring they've had the best experience of their life - Created engaging and accommodating games to campers of all ages and hockey levels.	

Bill Bolton Arena

Ice Hockey Referee

Toronto, ON

Part Time Oct - Mar 2022 - 2025

- Enforced game rules and safety policies while resolving on-ice disputes and promoting sportsmanship among youth players.

TECHNICAL SKILLS

Programming: C++, Python, Assembly

Embedded Systems: SPI, I²C, UART, Timers, EEPROM, Flash

PCB Design: Altium Designer, EasyEDA

Hardware & Test Equipment: Soldering, Oscilloscopes, Spectrum Analyzers, Multimeters

CAD: Fusion 360 (3D CAD)

Development Tools: Git, Visual Studio Code, Cursor

Productivity: Microsoft Excel, Microsoft Word